Vishal Chaudhary

M.Sc. Automation & Robotics

Research Interests: Geospatial Data Engineering, Digital Twins (UrbanTwin), Automated ETL Pipelines, and the integration of GIS data into Scalable DataLakes for AI Workflows.

Address	II. Bickestraße 21, 44263 Dortmund, Germany
Mobile	+49 17634667272
E-Mail	chaudharypinku1992@gmail.com
Date of Birth	01 April 1998
LinkedIn	www.linkedin.com/in/vishal-chaudhary-8b0978141
Github	https://vishalchaudharyprojects.github.io/

EXPERIENCE

08.2023 – Present

Research Assistant

IE3, TU Dortmund University, Dortmund, Germany

- Digital Twin & Data Standardization:
  - **CIM/CGMES Framework:** Engineered a Geospatial ETL/ELT Pipeline in Python to automate the ingestion and transformation of GIS XML data into standardized models. Implemented automated Data Cleansing and topological validation logic.
  - **Cloud-Native Microservices:** Architected a Scalable DataLake Architecture using Docker/FastAPI and S3 (Simple Storage Service) to manage high-volume grid models and metadata.
  - **Automated RDF/XML Generation:** Developed a high-fidelity mapping engine for grid entities (ACLineSegments, BatteryUnits, Transformers) ensuring strict compliance with IEC 61970.
- Hardware-in-the-Loop (HIL) & API Orchestration:
  - **Typhoon HIL Workflow API:** Architected a RESTful API (FastAPI) to orchestrate the simulation lifecycle, automating the transition from CIM data to compiled real-time execution on Typhoon HIL hardware.
  - **Asynchronous Task Management:** Implemented a job-tracking engine using UUID-based task management for parallel orchestration of model creation, compilation, and execution.
  - **Signal Mapping & Hardware Sync:** Developed a programmable signal mapping system for analog I/O blocks, including dynamic scaling to synchronize real-world hardware measurements.
- Secure Communication & Grid Analytics:
  - **Secure Hybrid Communication Engine:** Developed a multiprocessing ingestion tool for TLS-secured DLMS/COSEM and Modbus TCP, mapping raw payloads to IEC 61850 Logical Nodes.
  - **CIM-Based Automated Grid Analytics Pipeline:** Built an end-to-end toolchain for CIM 3.0 ingestion to automate Power Flow and WLS State Estimation via PandaPower, broadcasting results to InfluxDB and RabbitMQ for real-time analysis of networks.
  - **IEC 61850 & Modbus Integration:** Engineered a parser to extract parameters from IEC 61850 SCD files, automatically generating Modbus-TCP polling configurations and instantiating Logical Nodes for HIL validation.

EDUCATION

10.2021 - 12.2025

Master's degree in Automation and Robotics (M.Sc.)
TU Dortmund University, Dortmund, Germany

- **CGPA**: 2.3/4
- **Master Thesis:** Definition and Development of a Low Voltage Grid Digital Twin using the Asset Administration Shell as a Single Source of Truth for Power System Data and Services
  - **Key Contributions:** Developed a unified data model bridging GIS (topology), asset management data, and SCADA (real-time) data using IEC 63278.
  - **Outcome:** Currently first-authoring a research paper for a conference/journal submission in 2026 based on these findings.
- **Master Group Project:** Transparent Energy Transition - Generation of grid models for the digitization of "blind" low-voltage grids
  - Integrated group-generated GIS XML data with real-time streams from a Typhoon HIL model to engineer compliant CIM CGMES grid models, enabling precise powerflow analysis.
- **Academic Project:** Unsupervised Learning of Parametric Optimization Problems
  - Developed a robust Data Ingestion and Pre-processing pipeline for unsupervised neural network training, focusing on feature engineering and parametric optimization.

08.2015 - 06.2019

Bachelor of Technology: Mechanical Engineering
GLA University, Uttar Pradesh, India

- **CGPA**: 8.67/10
- **Focus**: Control Theory, Embedded Systems
- **Bachelors Project**: Design and Implementation of a PID-based Flight Control System for 6-DOF Quadcopter Stability: Developed an embedded controller in C++/Arduino to stabilize 6-DOF flight dynamics using tuned PID loops, achieving consistent autonomous takeoff and hover.

TECHNICAL SKILLS

Power Systems	CIM/CGMES (61970/61968), IEC 61850 (SCL/SCD), AAS (IEC 63278), DLMS/COSEM, PandaPower, WLS State Estimation
Software/DevOps	Python (FastAPI, PyModbus), Docker, RabbitMQ, InfluxDB, Simple Storage Services (S3), SQL(PostGIS/PostgreSQL), C/C++, Typhoon HIL, Gurobi
Languages	English (C1), German (B1), Hindi (Native)

PUBLICATIONS

1. **[First Author - In Preparation]** "A Low Voltage Grid Digital Twin using the Asset Administration Shell as a Single Source of Truth," to be submitted to an international journal/conference in 2026.

08.04.2026, Dortmund Vishal Chaudhary